CS 105: Department Introductory Course on Discrete Structures

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Graph theory

Matchings!

Lecture 21
Oct 24 2025

Topic 4: Graph theory (Recap)

1. Basics: graphs, paths, cycles, walks, trails.
2. Eulerian graphs: characterization using degrees of vertices.
3. Bipartite graphs: characterization using odd length cycles.
4. Graph isomorphisms and automorphisms.
5. Subgraphs of a graph: Cliques and independent sets.
6. Connected components, cut-edges, charac via cycles.

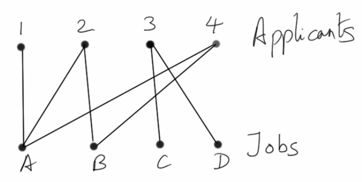
Reference: Most of Chapter 1 from Douglas West.

A General Recipe

1. Consider an "interesting" problem.
2. Model it as a "special" class of graphs.
3. Characterize them using properties on vertices, cycles etc.
4. Not done in this course: Build algorithms based on the characterization, analyze and implement them!

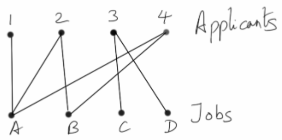
Matchings

- Suppose m people are applying for n different jobs. But not all applicants are qualified for all jobs, and each can hold at most one job.
- Then can you find a unique way to match jobs to applicants?



- What are the properties of such an assignment?
- Another practical example: the dating scene!

Matchings

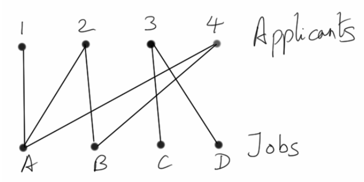


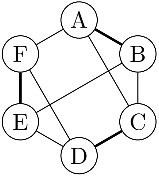
Definitions

- A matching in graph G is a set of (non-loop) edges having no shared end-points.
- The vertices incident to edges in a matching are called matched or saturated. Others are unsaturated.
- A perfect matching in a graph is a matching that saturates every vertex.

Is there a perfect matching in graph above? What if every applicant for qualified for every job?? How many perfect matchings are possibly in $K_{n,n}$?

Matchings in general graphs



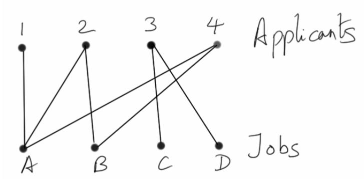


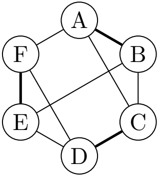
- A matching in a graph G is a set of (non-loop) edges with no shared end-points.
- The vertices incident to edges in a matching are called matched or saturated. Others are unsaturated.
- A perfect matching in a graph is a matching that saturates every vertex.
- What is such an example? Choosing roommates from a bunch of n people who may like or not like everyone.Or study-pairs...

Exercise: How many perfect matchings are possible in K_n?

- If even, i.e., for K_{2n}, no. of perfect matchings $f(n) = (2n-1)f(n-1) = \frac{(2n)!}{2^n n!}$.

Maximum matchings





Perfect matchings are not always possible; so what is the best we can do?

Definitions

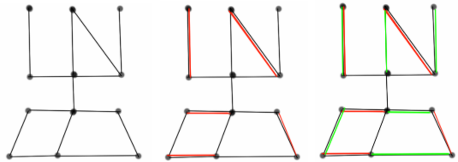
- A maximal matching in a graph is a matching that cannot be enlarged by adding an edge.
- A maximum matching is a matching of maximum size (in terms of number of edges) among all matchings in a graph.

- A matching is maximal if every edge not in it is incident to some edge in it.
- Does maximum $\implies$ maximal? Does maximal $\implies$ maximum?

Ex. Find smallest graph having a maximal matching that is not maximum. P_4.

Alternating and Augmenting paths

When do we know there must be a larger matching?



Definition

Given a matching M,

- an *M-alternating path* is a path that alternates between edges in M and edges not in M.
- An M-alternating path whose endpoints are unmatched by M is an *M-augmenting path*.

If a matching has an M-augmenting path, then can it be a maximum matching?

Characterization of Maximum matchings

- Clearly, a maximum matching cannot have an M-augmenting path.
- In fact, this is the characterization!

Theorem

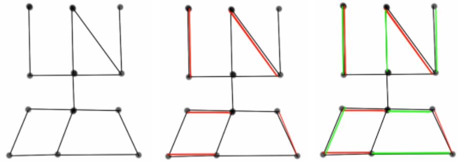
A matching M in G is a maximum matching iff G has no M-augmenting path.

Characterizing maximum matchings

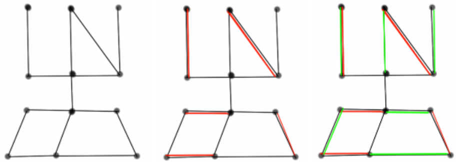
We need a definition and a lemma.

Definition

If M, M' are matchings in a graph H, the symmetric difference $M \triangle M'$ is the set of edges which are either in M or in M' but not both, i.e.,
$M \triangle M' = (M \setminus M') \cup (M' \setminus M)$.



Characterizing maximum matchings



Lemma

Every component of the symmetric difference of two matchings is either a path or an even cycle.

- Let $F = M \triangle M'$. F has at most 2 edges at each vertex, hence every component is a path or a cycle (why?).
- Further every path/cycle alternates between edges of $M \setminus M'$ and $M' \setminus M$.
- Thus, each cycle has even length with equal edges from M and M'. $\square$

Characterizing maximum matchings

Theorem (Berge'57)

A matching M in G is a maximum matching iff G has no M-augmenting path.

Proof:

- One direction is trivial (which one?!).
- ($\Longleftarrow$) For the other, we will show the contrapositive.
- i.e., if $\exists$ matching M' larger than M, we will construct an M-augmenting path.
- Let $F = M \triangle M'$. By Lemma, F has only paths and even cycles with equal no. of edges from M and M'.
- But then since $|M'| > |M|$ it must have a component with more edges in M' than M.
- This component can only be a path that starts and ends with an edge of M'; i.e., it is an M-augmenting path in G.